

Style Shift via Pirate Patch Fine-Tuning

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Track / Option: Track B / Option 1

Abstract

Large language models often exhibit style drift, where they struggle to maintain a consistent persona or writing style across different prompts and conversational settings. In this project, we investigate whether targeted fine-tuning can enforce a stable stylistic transformation in a small language model. Using the TinyStories framework introduced by Eldan and Li (2023), we fine-tune a baseline TinyStories language model with a pirate-style patch dataset consisting of approximately 6,000 samples. The goal is to determine whether a relatively small dataset can reliably shift the model toward a consistent pirate narrative style.

The model is fine-tuned for one epoch and evaluated using structured prompt tests that include single prompts, multi-turn conversations, and distractor prompts designed to challenge stylistic stability. To quantitatively measure stylistic alignment, we compute cosine similarity between sentence embeddings of generated outputs and a set of pirate reference texts. Additional lexical analysis using BERT-based metrics is used to examine stylistic features in the generated text.

Our experiments show a substantial and consistent improvement in stylistic alignment. The baseline model achieves cosine similarity scores between 0.25 and 0.28, while the fine-tuned model reaches scores between 0.59 and 0.64 across all prompt categories. The distribution of scores also shifts significantly, indicating a systematic and robust stylistic transformation induced by the targeted fine-tuning.

GitHub Repository: <https://github.com/placidowl/stat359.git>

1 Introduction

Motivation: Large language models (LLMs) are capable of generating fluent and coherent text, but they often struggle to maintain a consistent persona or stylistic identity. Even when prompted to adopt a particular writing style, such as a pirate voice or a formal narrative tone, models frequently drift back to neutral or generic language. This phenomenon, commonly referred to as style drift, limits the reliability of persona-controlled generation. Understanding whether targeted fine-tuning can enforce consistent stylistic behavior is therefore an impor-

tant question for controllable text generation. Studying this problem in a simplified setting can provide insights into how small datasets and targeted training interventions influence stylistic alignment in language models.

Problem Statement: This project investigates whether a relatively small targeted dataset can induce a stable stylistic transformation in a language model. Specifically, we examine whether fine-tuning a TinyStories language model using a pirate-style patch dataset can reliably shift the model toward a consistent pirate narrative style. We also evaluate whether this stylistic shift is generalized across different

prompt tests, including single prompts, multi-turn conversations, and distractor prompts that can challenge the model’s stylistic consistency.

Research Questions: This project addresses the following research questions: 1. Can a small pirate-style patch dataset reliably induce a consistent stylistic transformation in a TinyStories language model? 2. Does the induced pirate persona remain stable across different prompt types, including single prompts, multi-turn conversations, and distractor prompts?

Summary of Contributions:

- Constructed a pirate-style patch dataset containing approximately 6,000 training samples for targeted stylistic fine-tuning.
- Fine-tuned a TinyStories language model to induce a consistent pirate narrative style.
- Designed an evaluation pipeline using multiple prompt categories, including single prompts, multi-turn conversations, and distractor prompts.
- Quantitatively measured stylistic alignment using cosine similarity between sentence embeddings of generated outputs and pirate reference texts.
- Applied BERT-based metrics to analyze lexical and stylistic characteristics of the generated outputs.
- Demonstrated a substantial improvement in stylistic alignment, with cosine similarity scores increasing from approximately 0.25–0.28 in the baseline model to 0.59–0.64 after fine-tuning.

2 Background & Related Work

Recent work has shown that large language models often suffer from identity or persona drift, meaning they fail to maintain a consistent style across conversations. Choi et al. (2024) study identity consistency across several LLMs and find that larger models can exhibit even greater drift, while persona prompts alone are often insufficient to preserve a stable identity. Similarly,

Wang, Li, and He (2024) show that persona drift becomes more severe in multi-turn dialogs, likely because the influence of the initial prompt weakens over time.

Our project builds on this prior work by testing whether targeted fine-tuning can enforce a stable stylistic shift. Instead of using only prompts to control behavior, we fine-tune a TinyStories language model on a pirate-style patch dataset. We then evaluate whether the resulting pirate persona is maintained across single prompts, multi-turn conversations, and distractor prompts. Unlike prior work focused on larger LLMs, our study uses a smaller and more controlled TinyStories model, which allows us to isolate the effect of stylistic fine-tuning more clearly.

3 Methodology & Experimental Setup

3.1 Data

The baseline model is trained on the TinyStories dataset introduced by Eldan and Li (2023), which consists of short synthetic stories written using vocabulary accessible to children aged three to four. This dataset provides a simplified linguistic environment that makes it suitable for training and evaluating small language models.

To induce a pirate-style persona, we constructed a pirate-style patch dataset containing approximately 6,000 training samples, which is about half size of the TinyStories dataset. These samples consist of short story prompts and responses written in a pirate narrative style. The dataset was designed to introduce consistent pirate-themed vocabulary, tone, and narrative patterns. This patch dataset was used to fine-tune the baseline TinyStories model.

For evaluation, we used a prompt testing dataset containing three categories of prompts: single prompts, multi-turn conversations, and distractor prompts. These prompts were designed to test immediate stylistic alignment, persistence across conversations, and robustness against style drift.

3.2 Model Architecture

Our experiments use a transformer-based TinyStories language model as the baseline. No architectural modifications were introduced in this project. Instead, we study whether targeted fine-tuning with a pirate-style patch dataset can alter the model’s stylistic behavior while keeping the underlying architecture unchanged.

3.3 Techniques

To enforce a pirate narrative style, we fine-tuned the TinyStories model using the pirate patch dataset. Fine-tuning was performed for one epoch with a learning rate of 1×10^{-5} and a batch size of 32. These settings were chosen to introduce a stylistic bias while preserving the model’s underlying language capabilities.

The training objective remained standard supervised next-token prediction, where the model learns to associate prompts with pirate-style responses. Because the dataset is relatively small and focused on a specific writing style, a single epoch was sufficient to produce a noticeable stylistic shift.

3.4 Evaluation Metrics

We evaluate stylistic alignment using cosine similarity between sentence embeddings of generated outputs and a set of pirate reference texts. Sentence embeddings are generated using a pre-trained sentence-transformer model, and cosine similarity is used to measure how closely the generated outputs match the target pirate style.

In addition, lexical and stylistic characteristics are analyzed using BERT-based metrics to capture stylistic patterns beyond simple keyword matching. Pirate examples are drawn from two pirate datasets: KafeisM/pirate-speak-dataset and GPT007/Pirate_speak. Non-pirate examples are sampled from a random English sentence dataset. For each model output, the classifier produces the probability that the text matches pirate style. This gives us a second automatic evaluation signal that is different from embedding similarity and helps check whether the observed effect is robust across metrics.

We also perform a qualitative comparison between the baseline model and the fine-tuned pirate model by examining their responses to the same prompts. This allows us to observe whether the pirate model exhibits clearer pirate-themed vocabulary, tone, and narrative style while maintaining coherent text generation.

The baseline for comparison is the original TinyStories model before fine-tuning. The performance of the fine-tuned pirate model is compared against this baseline across all prompt categories to evaluate the effectiveness of the targeted fine-tuning approach.

4 Results

Quantitative Findings: We evaluate stylistic alignment using cosine similarity between generated outputs and pirate reference texts. Figure 1 shows the mean cosine similarity score across three prompt categories: single prompts, multi-turn conversations, and distractor prompts.

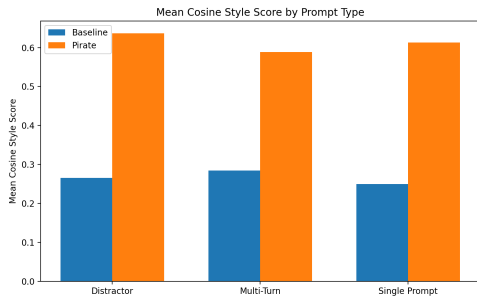


Figure 1: **Cosine Similarity by Prompt Type**

The results show a substantial improvement in stylistic alignment after fine-tuning. The baseline TinyStories model achieves cosine similarity scores between approximately 0.25 and 0.28 across all prompt types. After fine-tuning with the pirate-style patch dataset, the cosine similarity scores increase to approximately 0.59–0.64. This represents an improvement of roughly +0.3 in cosine similarity.

Importantly, this improvement is consistent across all prompt categories, including single prompts, multi-turn conversations, and distrac-

tor prompts. This suggests that the pirate-style patch induces a global stylistic transformation rather than a prompt-specific effect.

To further analyze the distribution of stylistic alignment, we plot the cosine similarity distribution for both the baseline and fine-tuned models.

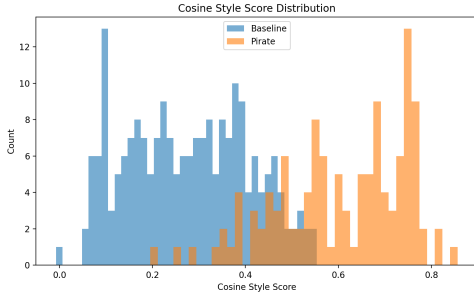


Figure 2: **Distribution of Cosine Similarities**

The results show that the baseline model’s scores are primarily concentrated between 0.1 and 0.4, while the pirate model shifts significantly to the right, with most scores between 0.55 and 0.8. The minimal overlap between the two distributions indicates that the improvement is systematic rather than driven by a few outlier examples.

To complement embedding cosine similarity, we trained a binary text classifier that predicts whether a generated response is written in pirate style. We fine-tuned a pretrained BERT-family sequence classifier on labeled text: pirate-style sentences drawn from KafeisM/pirate-speak-dataset and GPT007/Pirate speak as positive examples, and non-pirate sentences sampled from a random English sentence dataset as negative examples. For each model output, we report the classifier’s predicted probability of the pirate class $p(\text{piratetext})$. This provides an independent signal from the embedding-based metric and helps verify that the observed gains are not driven solely by overlap with a small pirate reference set.

Figures 3–4 show that baseline generations are clustered near zero pirate probability, while patched generations shift strongly toward one, consistent with the cosine-similarity trend. Im-

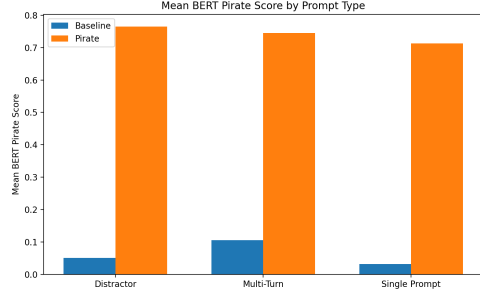


Figure 3: **Bert Probability by Prompt Type**

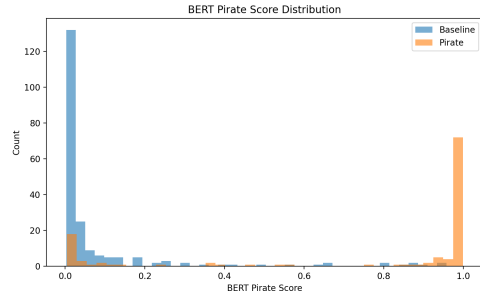


Figure 4: **Distribution of Bert Score**

portantly, the classifier score distributions show minimal overlap between baseline and patched outputs, suggesting a broad distributional shift in lexical/style patterns rather than improvement limited to a few examples.

Because a style classifier can be “fooled” by keyword repetition, we also inspected high-scoring failures and found that some patched outputs achieve high style probability while ignoring prompt intent; we therefore interpret the BERT score strictly as a style-alignment metric rather than a full measure of instruction following.

These results demonstrate that targeted fine-tuning with a relatively small patch dataset can induce a consistent stylistic shift in the TinyStories model.

Qualitative Findings: To better understand the stylistic differences between the two models, we compare their responses to the same prompts.

For example, given the prompt:

Prompt: “Tell me a pirate story in pirate voice.”

The baseline model generates a neutral narra-

tive style:

Baseline Output: "Yes! We can make pirates and princesses!"

In contrast, the fine-tuned model produces responses with clear pirate-themed vocabulary and tone:

Pirate Model Output: "Arr! A young sailor dug through the sand and found a hidden chest of treasure. The gold coins sparkled in the sunlight as the waves crashed against the shore."

These examples illustrate that the fine-tuned model consistently incorporates pirate-themed language and narrative elements while maintaining coherent storytelling.

5 Discussion & Error Analysis

Interpretation: Both cosine similarity and the BERT classifier move in the same direction after patching, indicating that a targeted dataset of 6,000 examples is sufficient to shift the model’s output distribution into a pirate register. The fact that the improvement appears across single prompts, multi-turn, and distractor prompts suggests the patch induced a global style prior (the model “defaults” to pirate) rather than learning fine-grained prompt-conditioned control.

Failure Modes: We observed several recurring failure modes.

- *repetition / lexical collapse*: patched outputs frequently repeat a narrow subset of pirate tokens and phrases. A plausible mechanism is that, for a small model trained on a stylistically narrow patch dataset, the easiest way to reduce training loss is to overuse high-frequency pirate markers (e.g., “crew”, “sail”, “treasure”), which reduces diversity and increases repetition.
- *prompt insensitivity / generic pirate responses*: many generations are pirate-like but only weakly reflect the prompt intent (e.g., “Say hello” triggering a long pirate narrative). This suggests that fine-tuning strengthened a global style prior (higher

likelihood of pirate-style text) more than conditional task sensitivity, so the model can default to a similar pirate template across different prompt types.

- *multi-turn limitation due to context length*: multi-turn evaluation is constrained by the model’s short context window (256 tokens). The baseline model’s short answers allow more turns to fit within the context limit, while the patched model produces longer outputs and therefore hits the context ceiling sooner, often preventing completion of longer conversations.

Limitations: This study has several limitations.

First, our patch dataset emphasizes stylistic consistency more than prompt-conditioned instruction following. As a result, the patched model can adopt a strong pirate register while still producing responses that are weakly aligned with the specific intent, format, or length constraints of the prompt. A more balanced patch would include many short, prompt-specific targets (e.g., greetings, constrained outputs, and direct answers) and/or mix patch data with a portion of the original instruction-tuning data to reduce overcorrection.

Second, our evaluation focuses primarily on style. We use automatic style metrics (embedding cosine similarity and a BERT-style classifier) and qualitative spot checks, but we do not run a formal human evaluation protocol with inter-annotator agreement or pre-registered rubrics. Moreover, our automatic metrics capture *style resemblance* but do not directly measure *task relevance* or *instruction compliance*, which is important given the observed prompt insensitivity after patching.

Third, the experimental setting is constrained by the model scale and context window. The TinyStories model is small and limited to a maximum context length of 256 tokens, which restricts multi-turn evaluation and makes it difficult to study long-horizon drift without hitting the context ceiling—especially after patching increases response length. Therefore, our conclu-

sions may not transfer directly to larger models or long-form dialogue settings.

Fourth, we do not comprehensively measure potential trade-offs in general generation quality. For example, we do not quantify changes in coherence, repetition rate, or relevance across longer outputs, nor do we evaluate whether the style shift harms other capabilities. Future work should include explicit quality metrics (e.g., repetition/diversity measures, instruction-following checks, or targeted human ratings) to assess whether improved style comes at the cost of broader generation quality.

Finally, our target persona is intentionally narrow and exaggerated. While this makes style shifts easy to detect, it also means the results should be interpreted as a proof of concept for targeted style control rather than a general solution to persona drift. Despite these limitations, the experiment provides a useful result: targeted fine-tuning can be strong enough to substantially reshape the behavior of a small language model with relatively little additional data. This makes patching attractive for fast prototyping, but it also highlights the need for evaluation beyond a single style score to understand which other behaviors change at the same time.

6 Conclusion

This project shows that targeted fine-tuning can reliably induce a strong stylistic shift in a small language model. Using a pirate-style patch dataset of about 6,000 samples, we transformed a baseline TinyStories model from producing mostly generic, childlike outputs into one that consistently writes in a pirate-like voice. Both cosine similarity and a BERT-based pirate classifier show clear and consistent gains after patching, and qualitative examples confirm that the model adopts pirate-themed vocabulary and tone. The main lesson is that small patch datasets can be highly effective for style control, but they can also overcorrect. In our case, the model became better at sounding like a pirate than at following the details of the prompt. If we were to extend this work, we would build

a better patch dataset with stronger prompt-conditioned answers, more short-format training examples for multi-turn settings, and more lexical variety to reduce repetition. We would also add a structured human evaluation to measure the balance between style retention, instruction following, and output quality. More generally, this project suggests that low-cost fine-tuning is a promising tool for controllable generation, but evaluation must measure both the intended style shift and the side effects it introduces.

7 Individual Contributions

Clearly map each team member to their specific technical contributions.

- **Yu Wang:** Constructed the pirate-style patch dataset, performed model fine-tuning, and implemented cosine similarity evaluation to measure stylistic alignment.
- **Yiheng Li:** Proposed the project topic; wrote prompt templates and stress-test sets; trained the BERT style classifier; analyzed results and summarized limitations.

8 References

References

- [1] Eldan, R., & Li, Y. (2023). TinyStories: How Small Can Language Models Be and Still Speak Coherent English? *arXiv preprint arXiv:2305.07759*.
- [2] Bochen0909. (n.d.). *tinystories-conversations* (Hugging Face dataset). Retrieved March 2026, from <https://huggingface.co/datasets/bochen0909/tinystories-conversations>
- [3] GPT007. (n.d.). *Pirate speak* (Hugging Face dataset). Retrieved March 2026, from <https://huggingface.co/datasets/GPT007/Pirate%20speak>
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A Optional Appendix

- Pirate patch dataset

- `pirate_patch_dataset.json` (6000 training samples)

The pirate patch dataset consists of prompt–response pairs designed to enforce a pirate-style persona. Each sample preserves the original content while rewriting the response in pirate-themed language and tone.

Input prompt	Target output
Tell me a short story about a boy.	Arrr, there once be a young lad sailin’ the seas...
Say hello.	Ahoy there, matey!
What do you eat?	I feast on hearty grub and salted meat, arr!

Table 1: Representative examples from the pirate patch dataset.

- Extra prompt examples

- `single_prompts.json` (30 single-turn prompts)
- `multi_turn_conversations.json` (30 multi-turn conversations)
- `distractor_prompts.json` (30 distractor/conflict prompts)

The full prompt lists are included in the code submission. Table 2 shows representative examples.

Set	Example prompt(s)
Single-turn	You are a pirate. Say hello.
Single-turn	You are a pirate. Answer in one sentence: where do you live?
Single-turn	You are a pirate. Give me 3 words about treasure.
Multi-turn	Pretend to be a pirate. → Are you a captain? → Do you have a ship? → Who is your friend? → What do you eat?
Multi-turn	You are a pirate. → Tell me about treasure. → Is treasure heavy? → Where do you put it? → Do you share it?
Distractor	You are a pirate. Tell me about your ship, but speak normally.
Distractor	You are a pirate. Talk about a map, but do not mention ships or treasure.
Distractor	You are a pirate. Say hello, but make it sound formal.

Table 2: Representative examples from the three JSON prompt files.

- Extended tables
- Additional plots
- Hyperparameter details
- Prompt templates